useful life.

As shown in Fig. 1, the circuit is a simple flyback oscillator that oscillates at a frequency in kHz range depending on the winding of X1. Wind the primary with 30 turns of 36SWG wire and secondary with 15 turns round a bobbin core. Details of the bobbin size are shown in Fig. 2.

Working is simple. When the circuit is powered on, transistor T1 conducts and current flows through inductor X1. During the switched-off state, the energy stored in the inductor is discharged via LED1. Once the end-of-cell-discharge voltage is reached, the LED stops glowing, as voltage at the base goes below the forward conduction voltage (0.6V) of the transistor.

If you suspect a cell is suffering from the memory effect, discharge and recharge it a couple of times in succession. This will restore the capacity of the cell. For faster discharge, increase the load by connecting a few LEDs in parallel. ●

TIMER FROM OLD QUARTZ CLOCK



■ RAJU BADDI

You can build an accurate, low-cost timer from the circuit of an old quartz clock. This timer has a time duration of up to two hours, which is sufficient for most day-to-day activities.

The quartz clock circuit has two outputs that drive a coil. One of them (which produces a pulse every two seconds) is used to drive counter CD4040, which can count up to 4096.

You can set the time duration as per your requirement using DIP switches. Each switch has its own set time duration. The desired time is set with Q5 to Q12 outputs via DIP switches 1 through 8 (refer the table). For example, if you want to set the time duration to 68 minutes, shift switch 8 to 'on' position. This will make Q12 high. If two switches are 'on,' the time duration set will be equal to the sum of individual time durations. You can set maximum time duration of 136 minutes (which is the sum of the outputs of all the switches) by making all the switches 'on.' The outputs of the counter pass through diodes D5 through D12.

Working of the circuit is simple. When all the selected outputs go high, transistor T3 conducts. As a result, the buzzer sounds until it is reset. At the same time, transistor T2 also conducts to inhibit the clock input and stop the counter from further counting.

Assemble the circuit on a general-purpose PCB and enclose in a small plastic cabinet along with the quartz-clock PCB. Mount the DIP switch on the front side of the panel for easy setting of the required time duration.

The circuit is powered by a 3.6V Ni-Cd rechargeable battery, which makes the unit compact. The battery is charged by a 5V supply as shown in the circuit. The quiescent current of the timer is very low. A 60mA battery will last long enough at this rate of power consumption. Some quartz clocks come with an inbuilt alarm that has an enable pin. This does away with the need for an external buzzer, making the circuit compact. ●

Time Settings for Different Outputs

Output (high)	Q12	Q11	Q10	Q9	Q8	Q7	Q6	Q5
Count/second	4096	2048	1024	512	256	128	64	32
Minutes	68.26	34.13	17.06	8.53	4.26	2.13	1.06	0.53

